A note on data analysis using Bayesian inference

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March 21, 2026

Abstract

This notes provides a short introduction to Bayesian inference for data analysis, with emphasis on parameter estimation and model comparison using Markov Chain Monte Carlo (MCMC) methods. We review the fundamentals of Bayes' theorem, prior and posterior distributions, and the interpretation of credible intervals. Practical aspects of MCMC sampling, including convergence diagnostics and the construction of corner plots for visualizing multi-dimensional posteriors, are illustrated through examples. Additionally, we discuss model selection criteria such as AIC, AICc, and BIC, and their relationship to the chi-square goodness-of-fit. The material is intended as a self-contained guide for students and researchers seeking to apply Bayesian techniques in their own work.

Contents

1 Introduction **2**

2 Bayesian Inference Basics **2**

3 Markov Chain Monte Carlo (MCMC) **3**

4 Parameter Estimation and Summaries **4**
4.1 Corner Plots 4
4.2 Chi-Square Scatter Plots 4

5 Model Comparison **4**
5.1 Relationship Between χ^2 and Log-Likelihood 4
5.2 Akaike Information Criterion (AIC) 5
5.3 Bayesian Information Criterion (BIC) 5
5.4 Interpreting ΔAIC and ΔBIC 5
5.5 Chi-Square per Degree of Freedom 6

6 Worked Example: Fitting a Polynomial **6**
6.1 Linear Model (Degree 1) 6
6.1.1 Convergence Diagnostics 6
6.1.2 Posterior Distributions 7
6.1.3 Best-Fit Lines 8
6.1.4 Chi-Square Scatter Plots 8
6.2 Quadratic Model (Degree 2) 11
6.2.1 Convergence Diagnostics 11
6.2.2 Posterior Distributions 11
6.2.3 Best-Fit Lines 12
6.2.4 Chi-Square Scatter Plots 12
6.3 Model Comparison 16

7 Conclusion **16**

A List of Symbols **17**

1 Introduction

In many scientific disciplines, we are faced with the problem of estimating the parameters of a model given some observed data, as well as the problem of finding the most appropriate model which describes better our observed data. Classical statistics often relies on maximum likelihood estimation (MLE) and frequentist confidence intervals. However, a more powerful and intuitive framework is provided by Bayesian inference, which treats parameters as random variables and updates our knowledge using Bayes' theorem.

In this chapter, we will explore the fundamentals of Bayesian parameter estimation using Markov Chain Monte Carlo (MCMC) methods. We will learn how to obtain posterior distributions, summarize them with means and credible intervals, and visualize them using corner plots. Furthermore, we will discuss how to compare different models using information criteria such as AIC, AICc, and BIC, as well as the chi-square goodness-of-fit. For further reading, you can find relevant references to the discussion material in previous works, such as [1, 2, 3, 4, 5] and references therein.

2 Bayesian Inference Basics

Bayes' theorem states that the posterior distribution of a set of parameters $\boldsymbol{\theta}$ given data D is proportional to the likelihood times the prior:

$$p(\boldsymbol{\theta} \mid D) = \frac{p(D \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{p(D)} \propto p(D \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}).$$

Here:

- $\boldsymbol{\theta}$ denotes the unknown parameters of the model (e.g., slope and intercept in a linear fit).
- D represents the observed data.
- $p(\boldsymbol{\theta} \mid D)$ is the **posterior** probability distribution, which quantifies our updated knowledge about the parameters after seeing the data.

- $p(D \mid \boldsymbol{\theta})$ is the **likelihood**, i.e., the probability of observing the data given a specific set of parameters.
- $p(\boldsymbol{\theta})$ is the **prior** distribution, encoding our knowledge about the parameters before seeing the data.
- $p(D)$ is the **marginal likelihood** (or evidence), a normalisation constant that ensures the posterior integrates to 1.

The likelihood $p(D \mid \boldsymbol{\theta})$ measures how probable the data are for a given parameter set. For independent Gaussian errors, the likelihood is

$$p(D \mid \boldsymbol{\theta}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{(y_i - f(x_i; \boldsymbol{\theta}))^2}{2\sigma_i^2}\right],$$

where:

- n is the number of data points.
- y_i are the observed measurements (with uncertainties σ_i).
- x_i are the independent variables (e.g., positions).
- $f(x_i; \boldsymbol{\theta})$ is the model prediction (e.g., a polynomial).
- σ_i are the standard deviations of the measurement errors (assumed known).

Equivalently, we often work with the log-likelihood

$$\log p(D \mid \boldsymbol{\theta}) = -\frac{1}{2} \sum_{i=1}^{n} \left[\frac{(y_i - f_i)^2}{\sigma_i^2} + \log(2\pi\sigma_i^2)\right].$$

Minimizing the sum of squared residuals weighted by the inverse covariance matrix is equivalent to maximizing the log-likelihood. This sum is the chi-square statistic:

$$\chi^2(\boldsymbol{\theta}) = \sum_{i=1}^{n} \frac{(y_i - f(x_i; \boldsymbol{\theta}))^2}{\sigma_i^2} = \mathbf{r}^T \mathbf{C}^{-1} \mathbf{r},$$

where $\mathbf{r}(\theta) = \mathbf{data} - \mathbf{model}(\theta)$ is the residual vector and $\mathbf{C}$ is the data covariance matrix. The chi-square is often used as a measure of goodness-of-fit.

The prior $p(\boldsymbol{\theta})$ encodes our knowledge before seeing the data. It can be informative (e.g., a Gaussian with a specified mean and variance) or weakly informative (e.g., a uniform distribution over a wide range). The choice of prior influences the posterior, especially when data are scarce.

3 Markov Chain Monte Carlo (MCMC)

For most realistic models, the posterior cannot be computed analytically. MCMC methods allow us to draw samples from the posterior distribution. The idea is to construct a Markov chain whose stationary distribution is the posterior. After a burn-in period, the chain's states are samples from the posterior.

One of the most popular MCMC algorithms is the Metropolis-Hastings algorithm, but modern probabilistic programming languages like PyMC (which we use in our accompanying Python code) implement more efficient samplers such as Hamiltonian Monte Carlo (HMC) and the No-U-Turn Sampler (NUTS).

Convergence of the chains must be checked. A common diagnostic is the potential scale reduction factor $\hat{R}$ (Gelman-Rubin statistic), which compares the variance

within chains to the variance between chains. Values close to 1 (typically $\hat{R} < 1.01$) indicate convergence.

4 Parameter Estimation and Summaries

Once we have a set of posterior samples $\{\boldsymbol{\theta}^{(s)}\}_{s=1}^{S}$, we can estimate:

- **Posterior mean**: $\bar{\boldsymbol{\theta}} = \frac{1}{S}\sum_{s=1}^{S} \boldsymbol{\theta}^{(s)}$.
- **Posterior standard deviation**: a measure of uncertainty, often reported as 1σ error bars.
- **Credible intervals**: e.g., the 68% interval ($[p_{16}, p_{84}]$) and the 95% interval ($[p_{2.5}, p_{97.5}]$), obtained from the percentiles of the samples.

If the posterior is approximately Gaussian, the mean and standard deviation fully characterize it, and the 68% interval coincides with $\bar{\theta} \pm \sigma$.

We also often locate the maximum a posteriori (MAP) estimate or the minimum chi-square point, which corresponds to the mode of the posterior (with flat priors, this is the MLE).

4.1 Corner Plots

A corner plot (or triangle plot) is a powerful visualization tool. It shows the marginal histograms for each parameter on the diagonal and scatter plots or contour lines for each pair of parameters off-diagonal. Contours typically enclose 1σ and 2σ regions (39% and 86% probability for a 2D Gaussian). Such plots allow us to see correlations and the shape of the posterior.

4.2 Chi-Square Scatter Plots

For each parameter, we can plot χ^2 versus the parameter value for all posterior samples. The minimum χ^2 gives the best-fit point. Horizontal lines at $\chi^2_{\min} + \Delta$ (with $\Delta = 1$ for 1σ, $\Delta = 4$ for 2σ in the case of two parameters of interest) help identify confidence regions. For a single parameter, $\Delta\chi^2 = 1$ corresponds to a 68% interval if the profile likelihood is approximately quadratic.

5 Model Comparison

Often we have several competing models (e.g., a straight line vs. a quadratic). We need to decide which model is favored by the data, taking into account the complexity of the model (number of parameters). Several information criteria are commonly used.

5.1 Relationship Between χ^2 and Log-Likelihood

For independent Gaussian errors with known standard deviations σ_i, the log-likelihood is

$$\ln L(\boldsymbol{\theta}) = -\frac{1}{2}\chi^2(\boldsymbol{\theta}) - \frac{1}{2}\sum_{i=1}^{n} \ln(2\pi\sigma_i^2).$$

Thus, the maximized log-likelihood is

$$\ln \hat{L} = -\frac{1}{2}\chi^2_{\min} + \text{constant},$$

where the constant does not depend on the model parameters. In model comparison, this constant cancels when computing differences in AIC or BIC, so we can use $\chi^2_{\min}$ directly:

$$\text{AIC} = 2k - 2\ln \hat{L} = 2k + \chi^2_{\min} + \text{constant},$$
$$\text{BIC} = k\ln n - 2\ln \hat{L} = k\ln n + \chi^2_{\min} + \text{constant}.$$

Since the constant is the same for all models being compared, we simply compare $2k + \chi^2_{\min}$ and $k\ln n + \chi^2_{\min}$.

5.2 Akaike Information Criterion (AIC)

AIC estimates the relative amount of information lost by a model. It is defined as

$$\text{AIC} = 2k - 2\ln(\hat{L}) = 2k + \chi^2_{\min} + \text{constant}.$$

Lower AIC indicates a better model. For small sample sizes, a corrected version (AICc) is recommended:

$$\text{AICc} = \text{AIC} + \frac{2k(k+1)}{n-k-1}.$$

5.3 Bayesian Information Criterion (BIC)

BIC is another criterion derived from Bayesian model comparison:

$$\text{BIC} = k\ln(n) - 2\ln(\hat{L}) = k\ln(n) + \chi^2_{\min} + \text{constant}.$$

BIC penalizes complexity more strongly than AIC. It is consistent in the sense that as n grows, it will select the true model with probability approaching 1, under certain assumptions.

5.4 Interpreting ΔAIC and ΔBIC

When comparing models, we look at the absolute difference in AIC (or BIC). The following guidelines are commonly used for ΔAIC (the same qualitative interpretation applies to BIC):

$\lvert\Delta\text{AIC}\rvert \approx 0$–$2$	Models are essentially equivalent; no meaningful evidence to prefer one over the other.
$\lvert\Delta\text{AIC}\rvert \approx 2$–$4$	Weak support for the model with lower AIC.
$\lvert\Delta\text{AIC}\rvert \approx 4$–$10$	Substantial support for the lower-AIC model.
$\lvert\Delta\text{AIC}\rvert > 10$	Strong support for the lower-AIC model.

These thresholds are guidelines, not absolute rules.

5.5 Chi-Square per Degree of Freedom

A simple goodness-of-fit measure is the reduced chi-square:

$$\chi^2_\nu = \frac{\chi^2_{\min}}{\nu},$$

where $\nu = n - k$ is the number of degrees of freedom (n data points, k parameters). A value near 1 indicates a good fit, while values much larger than 1 suggest the model is too simple (underfitting), and values much less than 1 may indicate overfitting or overestimated errors.

6 Worked Example: Fitting a Polynomial

We illustrate the concepts with a synthetic dataset generated from a linear model:

$$y_i = 1.0 \cdot x_i + 0.0 + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, 0.3^2).$$

We consider two polynomial models: a straight line (degree 1) and a quadratic (degree 2). For each model, we run three inference scenarios:

1. **Prior only**: samples drawn from an informative Gaussian prior $p_0 \sim \mathcal{N}(0, 1.5)$, $p_1 \sim \mathcal{N}(1, 1.5)$ (and for quadratic, $p_2 \sim \mathcal{N}(0, 1.5)$).
2. **Data only**: flat uniform priors over $[-10, 10]$, so the posterior is essentially the likelihood.
3. **Data + Prior**: the same informative prior combined with the data.

All plots shown below were generated by the accompanying Python code using PyMC. The file names indicate the polynomial degree (e.g., `corner_plot_deg1.pdf` corresponds to the linear model).

6.1 Linear Model (Degree 1)

6.1.1 Convergence Diagnostics

Trace plots for the data-only and full (data+prior) models are shown in Figures 1 and 2. The $\hat{R}$ values are printed on the plots; all are below 1.01, indicating good convergence.

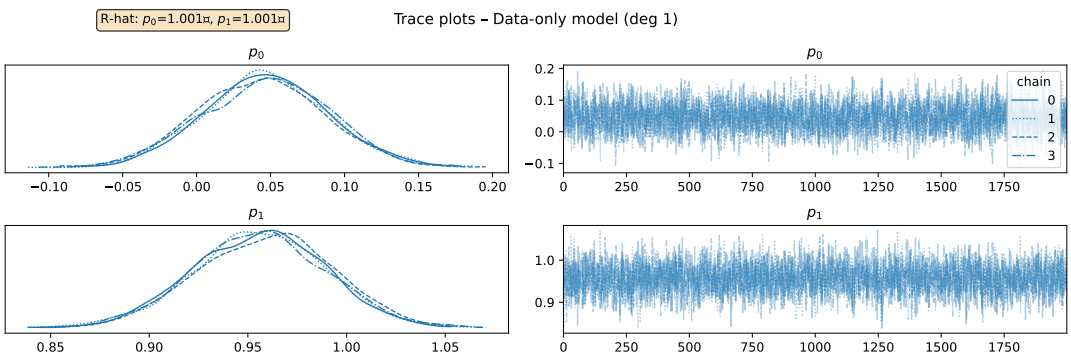

Figure 1: Trace plots for the linear model with flat priors (data-only). The $\hat{R}$ values confirm convergence.

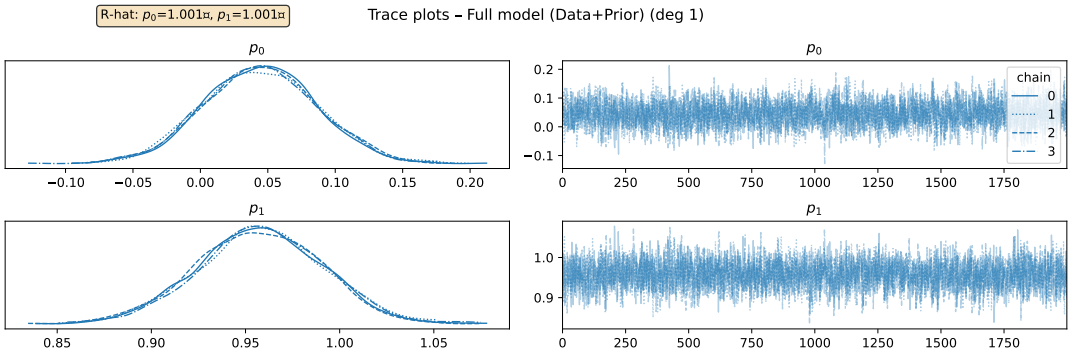

Figure 2: Trace plots for the linear model with informative priors (data+prior). Convergence is also achieved.

6.1.2 Posterior Distributions

Figure 3 shows the corner plot comparing the prior, data-only, and data+prior posterior distributions. The legend displays the mean $\pm 1\sigma$ for each parameter. The titles on the diagonal show the median and 68% intervals.

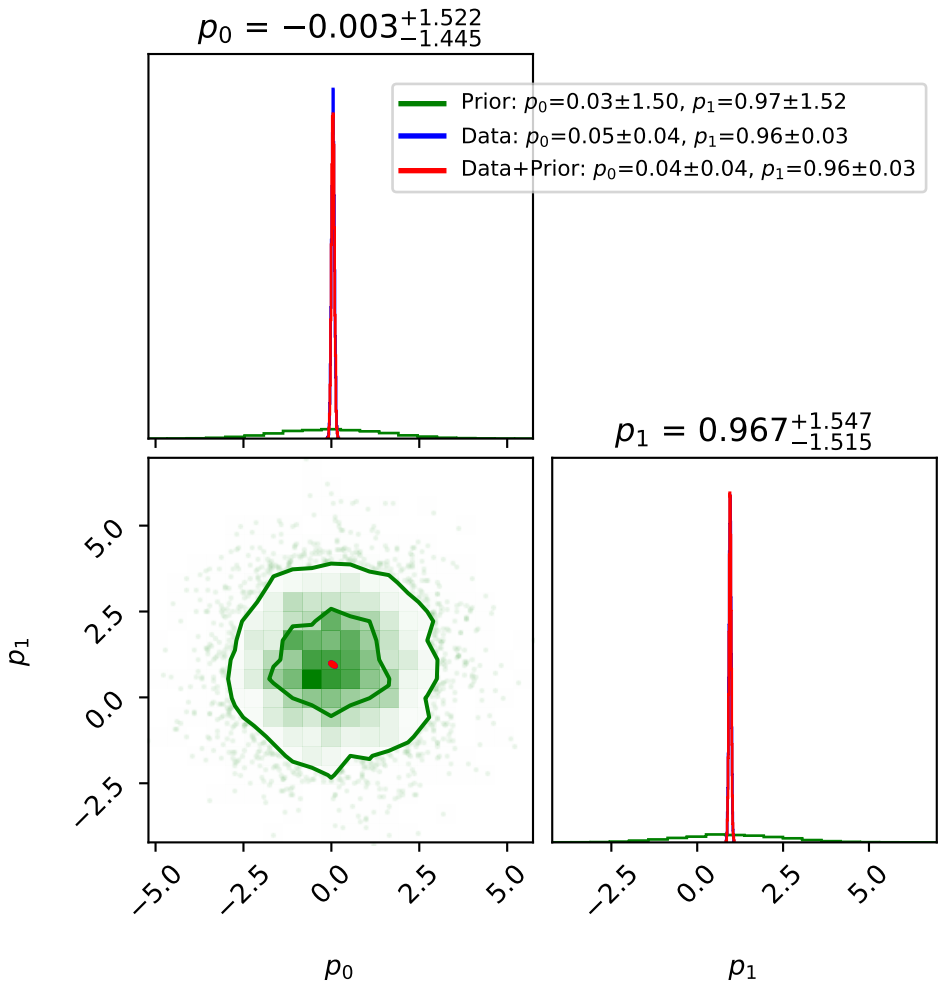

Figure 3: Corner plot for the linear model. The three cases are overlaid: prior (green), data-only (blue), and data+prior (red).

6.1.3 Best-Fit Lines

The best-fit models (using posterior means) are plotted against the data in Figure 4. The prior-only line is nearly flat, while the data-only and data+prior lines are almost identical, as expected when the prior is broad.

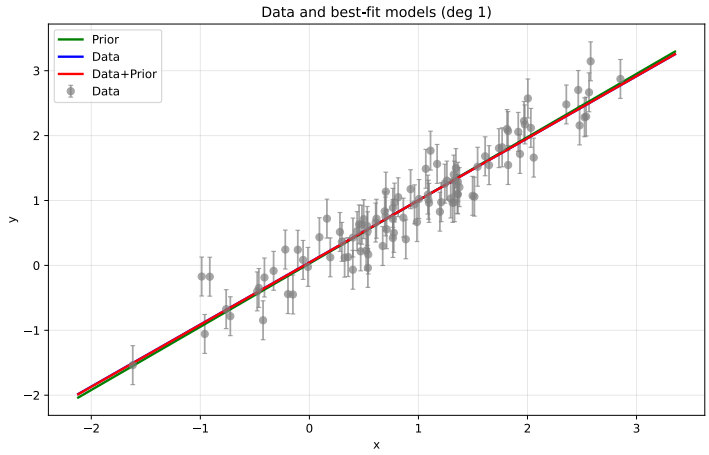

Figure 4: Data with error bars and the three best-fit lines for the linear model.

6.1.4 Chi-Square Scatter Plots

Figures 5, 6, and 7 show χ^2 versus each parameter for the prior, data-only, and data+prior cases. The minimum χ^2 is marked with a star, and horizontal lines indicate χ^2_{min}, $\chi^2_{\text{min}} + 1^2$, and $\chi^2_{\text{min}} + 2^2$. Text boxes give the 68% and 95% intervals from the posterior percentiles.

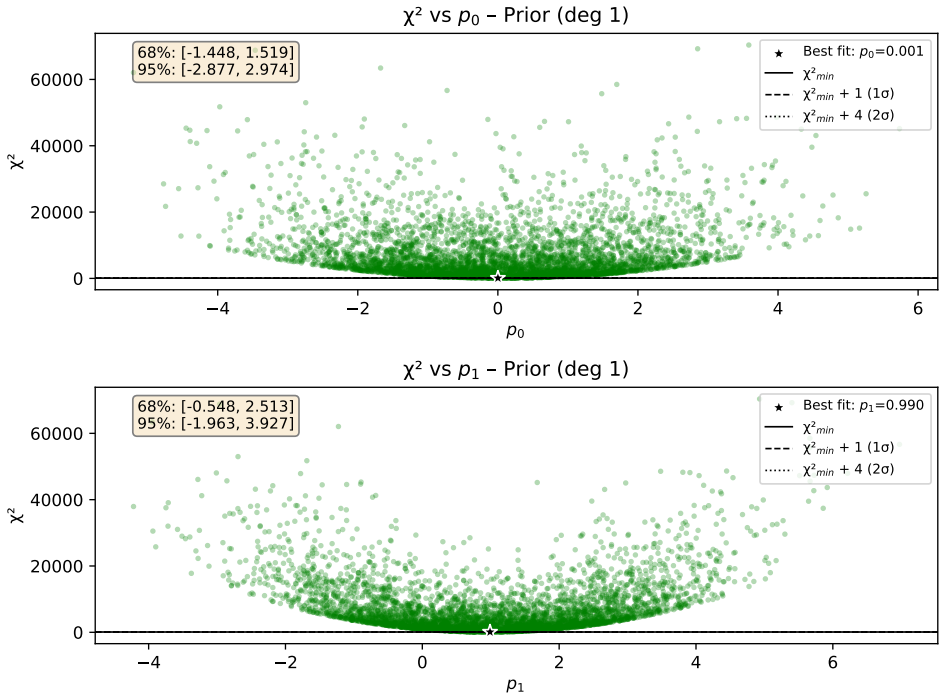

Figure 5: χ^2 vs. parameters for the prior-only case (linear model). Note, no data used here.

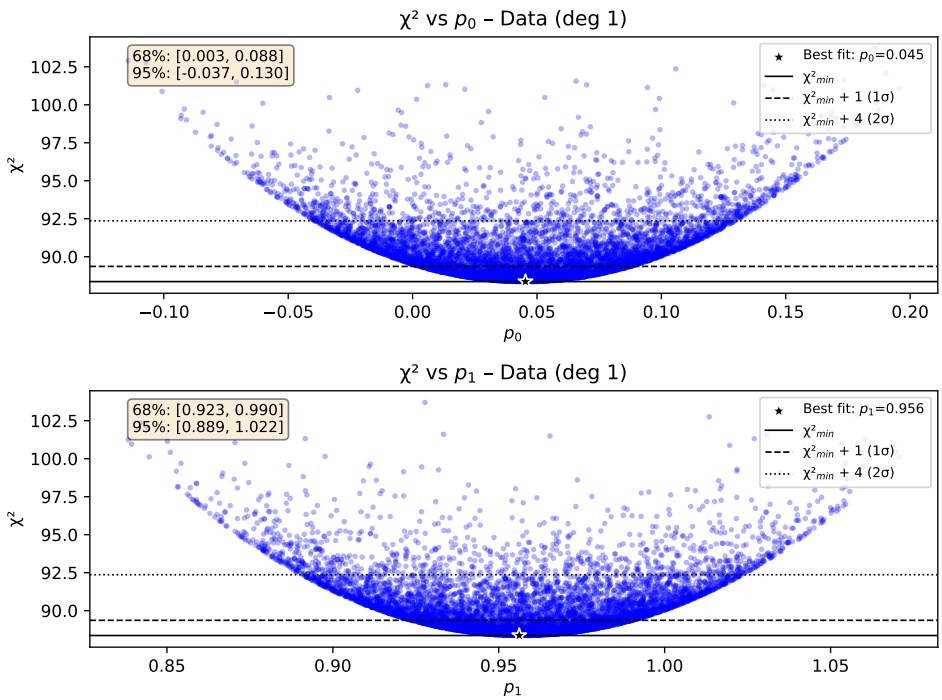

Figure 6: χ^2 vs. parameters for the data-only case (linear model).

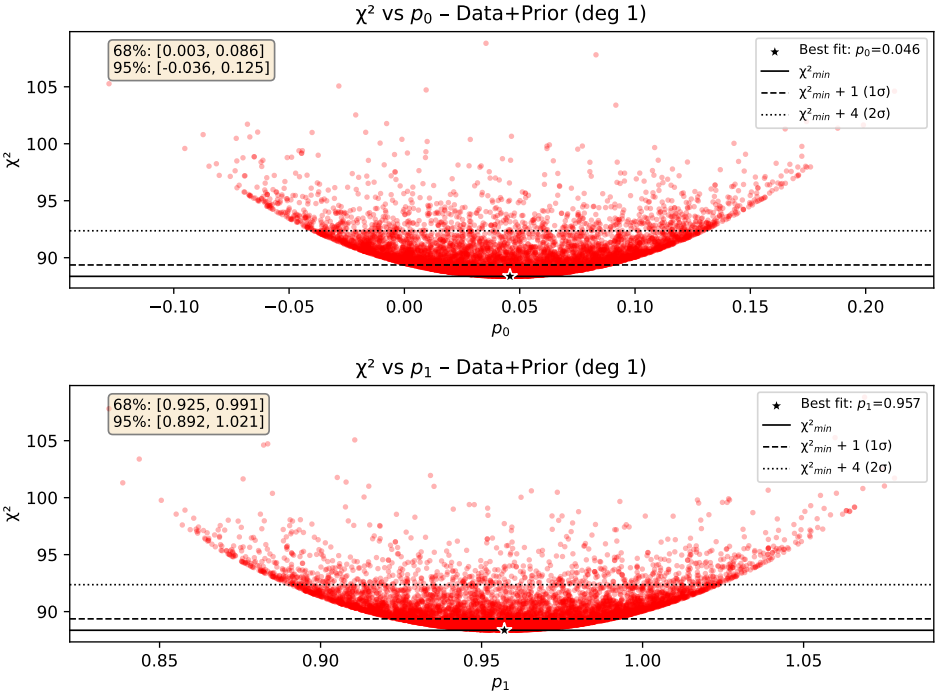

Figure 7: χ^2 vs. parameters for the data+prior case (linear model).

6.2 Quadratic Model (Degree 2)

We repeat the same analysis for a quadratic model. The data are still linear, so this model is unnecessarily complex. The plots below illustrate how the extra parameter behaves.

6.2.1 Convergence Diagnostics

Trace plots for the quadratic model (Figures 8 and 9) show good convergence.

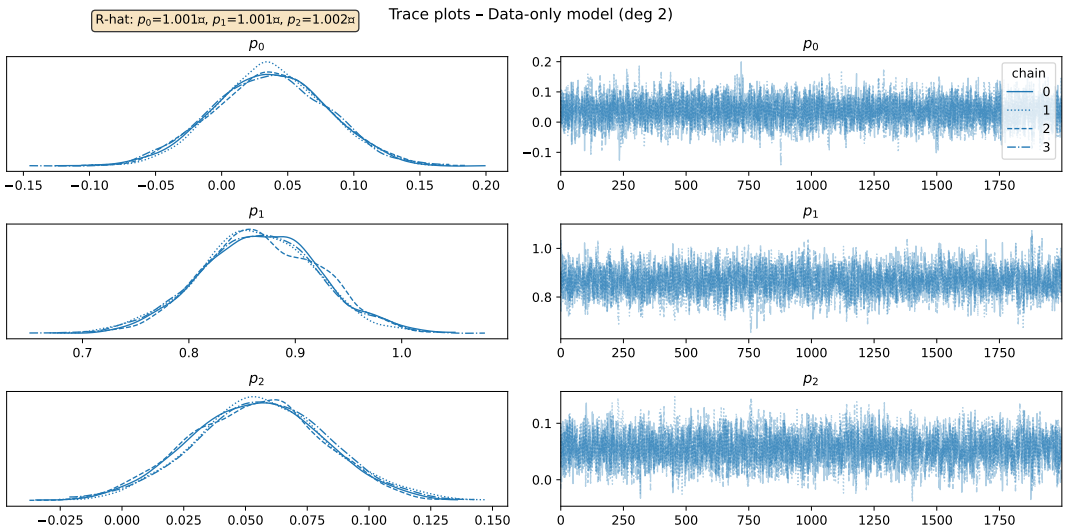

Figure 8: Trace plots for the quadratic model with flat priors (data-only).

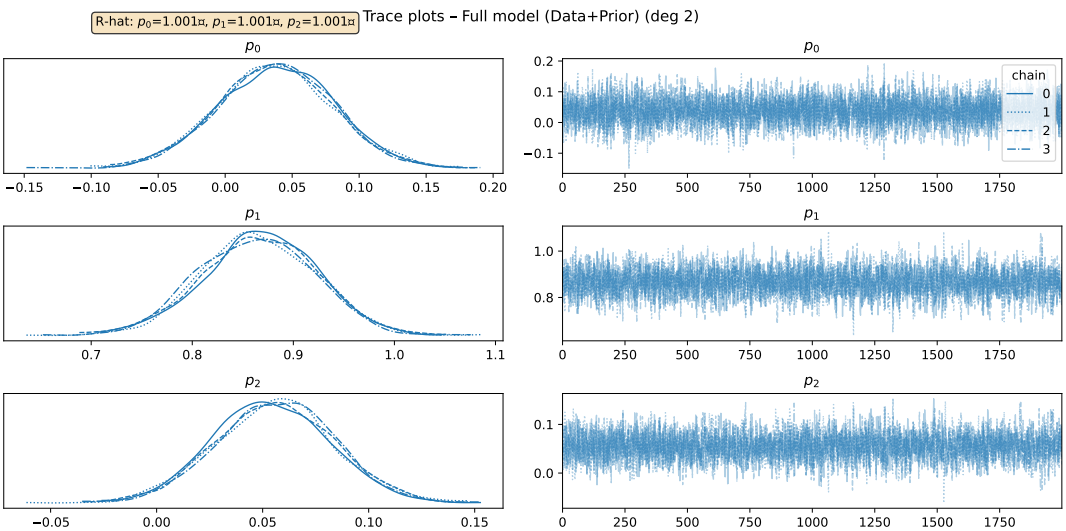

Figure 9: Trace plots for the quadratic model with informative priors (data+prior).

6.2.2 Posterior Distributions

The corner plot for the quadratic model (Figure 10) reveals that the extra parameter p_2 is consistent with zero, as expected.

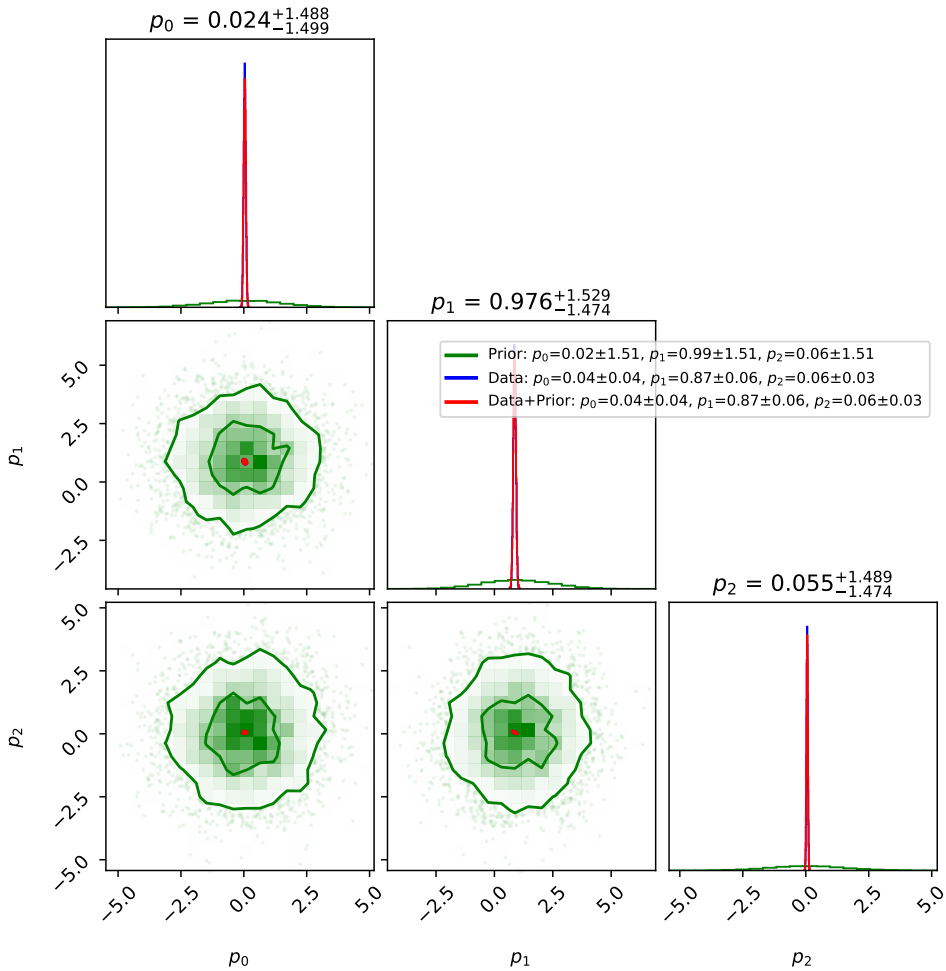

Figure 10: Corner plot for the quadratic model. The legend shows mean $\pm 1\sigma$ for all three parameters.

6.2.3 Best-Fit Lines

Figure 11 compares the three best-fit quadratic models. The prior-only line is nearly flat, while the data-only and data+prior lines show a slight curvature due to noise, but remain close to the true linear trend.

6.2.4 Chi-Square Scatter Plots

Figures 12 through 14 display χ^2 versus each of the three parameters for the three inference cases.

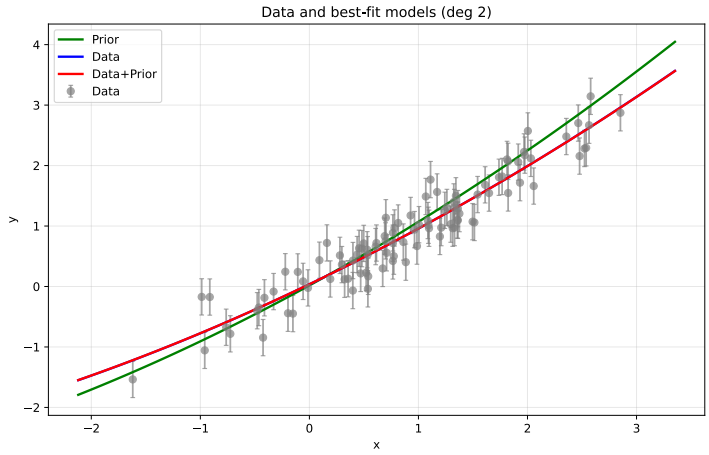

Figure 11: Data with error bars and the three best-fit quadratic models.

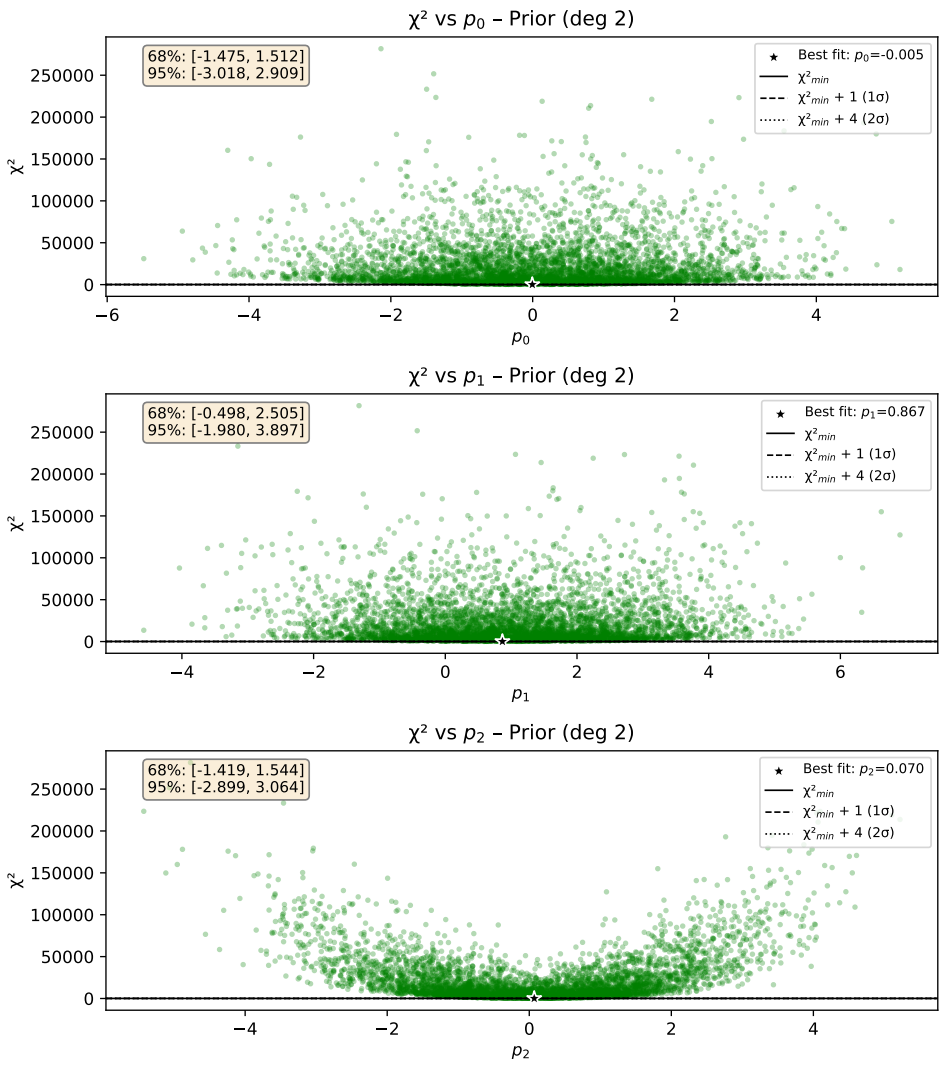

Figure 12: χ^2 vs. parameters for the prior-only case (quadratic model). Note, no data used here.

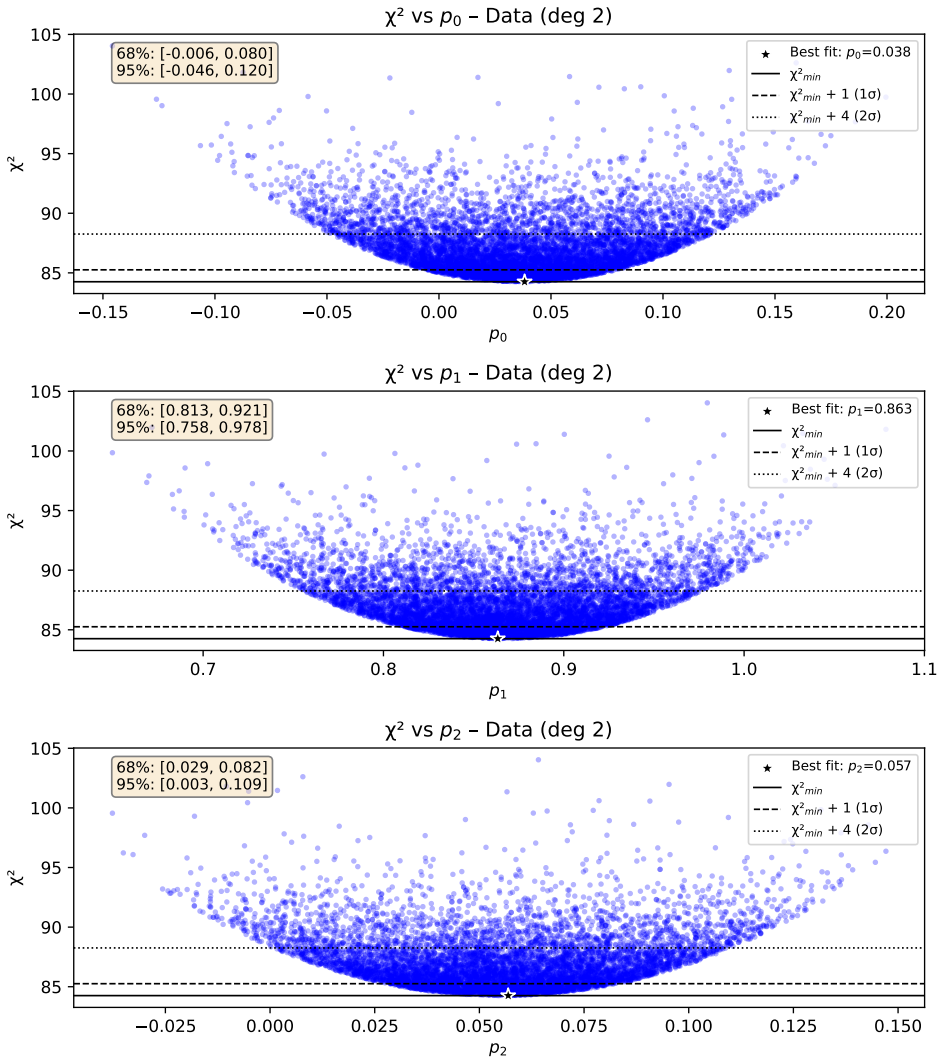

Figure 13: χ^2 vs. parameters for the data-only case (quadratic model).

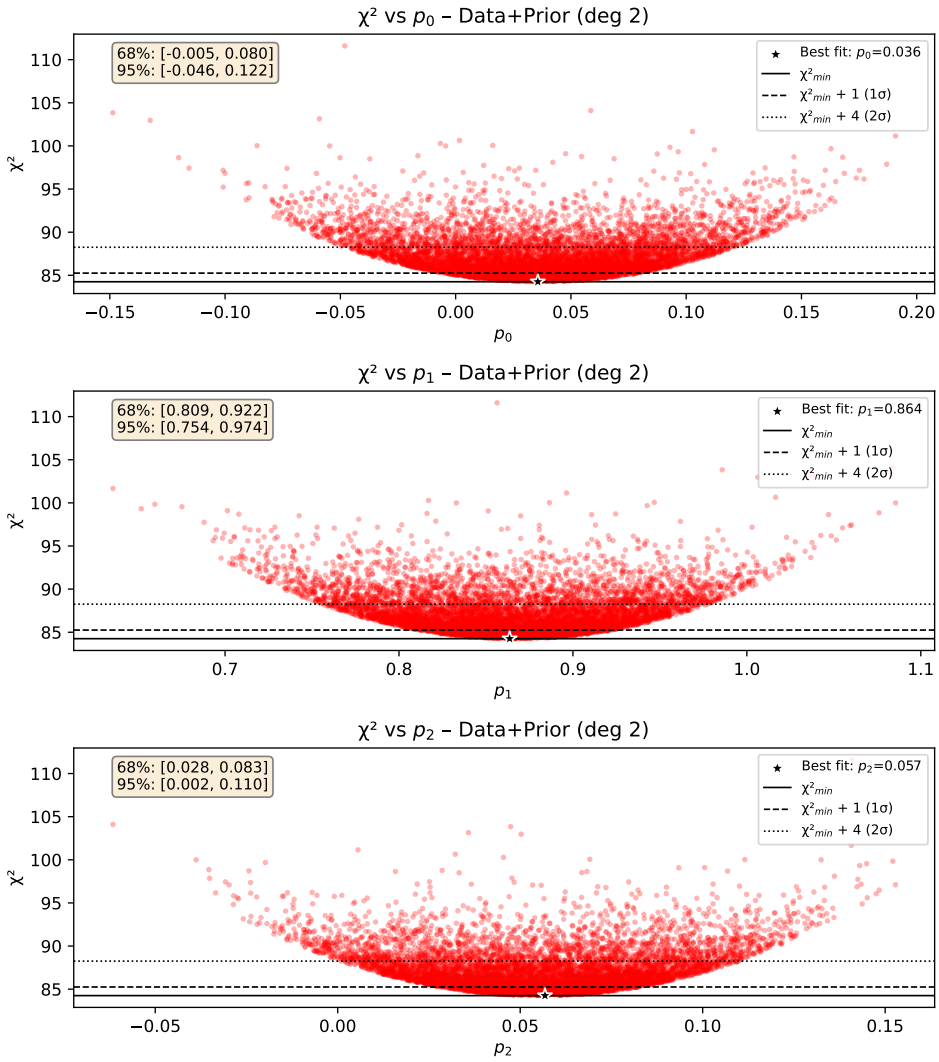

Figure 14: χ^2 vs. parameters for the data+prior case (quadratic model).

6.3 Model Comparison

We now quantitatively compare the linear and quadratic models using the information criteria discussed earlier. The results from the data-only fits are summarized in Table 1.

Table 1: Model comparison for linear ($k = 2$) vs. quadratic ($k = 3$) fit.

Model	k	$\chi^2_{\min}$	χ^2/ν	AIC	AICc	BIC
deg 1	2	88.36	0.902	92.36	92.49	97.57
deg 2	3	84.26	0.869	90.26	90.51	98.07

Interpretation:

- The reduced chi-square χ^2/ν is close to 1 for both models, indicating acceptable fits.
- AIC and AICc are slightly lower for the quadratic model (90.26 vs. 92.36). The difference $\Delta\text{AIC} = 2.1$ falls in the "weak support" range (2–4). Given that the data were generated from a linear model, this small difference is not surprising; the extra parameter does not harm the fit but also does not improve it enough to strongly favor the more complex model.
- BIC, which penalizes complexity more heavily, favors the linear model (97.57 vs. 98.07). The difference $\Delta\text{BIC} = 0.5$ is negligible, indicating the models are essentially equivalent.

Thus, based on these criteria, we might cautiously prefer the simpler linear model, in agreement with the true underlying process.

7 Conclusion

Bayesian inference with MCMC provides a flexible and powerful framework for parameter estimation and uncertainty quantification. Posterior samples allow us to compute means, standard deviations, credible intervals, and to visualize the joint distribution with corner plots. Chi-square analysis complements the Bayesian approach by offering a goodness-of-fit measure and helps identify confidence regions.

Model comparison using information criteria like AIC, AICc, and BIC enables us to select among competing models, balancing fit quality against model complexity. These tools are essential in modern data analysis across all scientific fields.

The accompanying Python code (using PyMC) implements the entire workflow, from data generation to MCMC sampling, visualization, and model comparison. It is designed to be easily adaptable to any polynomial model and can serve as a starting point for students and researchers.

Acknowledgements

PN acknowledges financial support from the National Natural Science Foundation of China (NSFC) under Grant No. U2541210.

A List of Symbols

The following table summarises the main symbols used in this document, along with their definitions and units where applicable.

Table 2: Symbols and their meanings.

Symbol	Description	Units / Remarks
$\boldsymbol{\theta}$	Vector of model parameters	varies (e.g., slope, intercept)
D	Observed data	depends on experiment
$p(\cdot)$	Probability density	
$p(\boldsymbol{\theta} \mid D)$	Posterior distribution	
$p(D \mid \boldsymbol{\theta})$	Likelihood function	
$p(\boldsymbol{\theta})$	Prior distribution	
$p(D)$	Marginal likelihood (evidence)	normalisation constant
$f(x; \boldsymbol{\theta})$	Model prediction	depends on x and $\boldsymbol{\theta}$
n	Number of data points	dimensionless
y_i	i-th measured value	same units as f
x_i	i-th independent variable	same units as x
σ_i	Standard deviation of measurement i	same units as y
$\mathbf{C}$	Covariance matrix of data	$\mathbf{C}_{ij} = \mathrm{Cov}(y_i, y_j)$
χ^2	Chi-square statistic	dimensionless
$\mathbf{r}$	Residual vector	$\mathbf{r} = \mathbf{y} - \mathbf{f}$
k	Number of model parameters	dimensionless
$\nu = n - k$	Degrees of freedom	dimensionless
χ^2_ν	Reduced chi-square	dimensionless
$\hat{R}$	Potential scale reduction factor	dimensionless, ≈ 1 indicates convergence
AIC	Akaike Information Criterion	dimensionless
AICc	Corrected AIC	dimensionless
BIC	Bayesian Information Criterion	dimensionless
ΔAIC	Difference in AIC	dimensionless
S	Number of MCMC samples	dimensionless

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